



NORTH SOUTH UNIVERSITY

Department of Biochemistry & Microbiology

Fall 2025

BIO 103: Basic Biology (Section: 38)
Department of Biochemistry & Microbiology (BMD)
School of Health and Life Sciences (SHLS)

Course instructor : Prof. Dr. Md. Rakibul Islam (MRIs)
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Office Room : SAC 823/1164
Office Hour : RA: 1:00 p.m. - 2:30 p.m.
Class Time : RA: 4:20 p.m. - 5:50 p.m.
Class Room : SAC 511

Disclaimer:

The instructor holds the right to make necessary changes to the syllabus and the grading policies outlined here to best accommodate the interest of the class.

If my scheduled office hours are not convenient for you, please make an appointment. The best way to reach me is by e-mail but feel free to talk with me after class or phone me at my office. If you are having problems in class, please see me as soon as possible. I will do everything I can to help you with your problems but don't wait until mid-semester or immediately before exams.

Rationale:

All university graduates are expected to have at least a basic knowledge of the core science courses. Students will need to take Biology I (BIO 103) to have a thorough understanding of the fundamental concepts in general biology of living organisms.

Course objective:

- Familiarize with the concept of chemistry of life and biological macromolecules associated with living systems.
- Understand the basic structural makeup of cells, how the organelles function to keep the cells functional and the generation of food and energy required for the survival of cells.
- Grasp the idea of the central dogma of life, understand how stored information get expressed, and how cellular division results in the continuity of life.
- Understand the processes involved in homeostasis in humans and get familiarized with human physiology.
- Get an overview of the diseases involved with different organ systems, food and nutrition, and diseases or disorders related to lifestyle.

Learning outcomes:

- By attending classes regularly, and through participation in the assessment exams (Class participation, Quizzes, Mid-Term, and Final), students of this course should be able to:
- Understand the basic concepts and principles of general biology.
- Recognize the chemical makeup of living organisms
- Identify and describe the structural components of the cell and their function.
- Understand and explore the importance of DNA in terms of regulating the function of living organisms, genetics, and biotechnology.
- Understand the chemical and physiological processes operating in the human body.
- Explore the knowledge of biology to be applied to health and lifestyle-related disorders.



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Course Timeline*

	Topics or Chapters	Assessment Items
Week 1 to 6	1. Introduction to Biology: Scope, Biology, what is life and Characteristics and classification of living things 2. Chemistry of life: Atoms & elements; Molecules & bonds; Electronegativity; Polar & non-polar bonds; Diffusion & osmosis; pH. 3. Biological Macromolecules: Carbohydrate, Lipid, Protein and Nucleic acids. 4. Central Dogma of life: Gene, genome, Flow of genetic information, Transcription and Translation. 5. Cell structure and function: Organelles description & level of organization.	Quiz 1, 2 & Midterm
Week 7 to 8	6. Energy of Life: Cellular respiration (anabolism & catabolism); Enzymes definition and characterization; Photosynthesis.	Quiz 3 & Final Examination
Week 8 to 12	7. Cellular Reproduction: Cell cycle, Mitosis, Meiosis. 8. Human Physiology: Homeostasis, digestive system, circulatory (blood), excretory and respiratory systems. 9. Health and Disease: Food and nutrition, Diabetes and lipid profile.	

*(The type of assessment items is subject to change and any such change will be notified to the student day ASAP)

Teaching-learning strategy:

(Dates and Times are subject to change and any such change will be notified to the student day before the class)

Lecture*	Lecture Topic	Time
1	Discuss the tentative lesson plan for the course	10 min
	Introduction to Biology (Part 1): Scope, definition, brief History, Modern biology, Unifying themes of biology, Life, Characteristics of living things: The need for energy, Movement, Cellular evolution, structure and organization,	70 min
	Interactive question-answer session	10 min
	Total	90 min
2	Recap and review on previous lecture	10 Min
	Introduction to Biology (Part 2): Growth and development, Maintenance and repair, Reproduction, Response to stimuli, Variation and adaptation, Metabolism); Hierarchical organization level of life with examples of Human and Plant.	70 min
	Interactive Q & A session	10 min
	Total	90 min
3	Recap and review on previous lecture	10 min



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	Chemistry of life (Part 1): Atoms & elements; Molecules & bonds; Electronegativity;	
	Q & A session	10 min
	Total	90 min
4-5	Recap and review on previous lecture	10*2 min
	Chemistry of life (Part 2): Polar & non-polar bonds, Hydrogen bond, Water: Properties, Characteristics, Distribution inside the human body; Solutions, Acids, Bases, pH, Biological Buffers; Diffusion, Osmosis, Active Transport	70*2 min
	Interactive Q & A session	10*2 min
	Total	90 min
6-7	Recap and review on previous lecture	10*2 min
	Biological Macromolecules (Part 1): Carbohydrate, Lipid, Protein and Nucleic acids: Compositional and functional features; Classes; Sources in nature; Examples; Occurrence as a component of organisms and nutritional.	70 *2 min
	Interactive Q & A session	10*2 min
	Total	90 min
8-9	Recap and review on previous lecture	10*2 min
	DNA and RNA: Genetic information of Life; Differences between DNA and RNA; Gene, Genome: Definition; Nuclear and Mitochondrial Genome.	70*2 min
	Interactive Q & A session	10*2 min
	Total	90*2 min
10	Recap and review on previous lecture	10 min
	Central Dogma of Molecular Biology: Flow of Genetic Information: Sketch the pathway of protein synthesis by mentioning the process involved; Transcription, Translation: Definition; where those occur (in which organelle).	70 min
	Interactive Q & A session	10 min
	Total	90 min
11	Recap and review on previous lecture	10 min
	Cell structure and function (Part 1): Cell: Definition, Cell Theory; Types of cells; Prokaryote, Eukaryote with examples; Common cell features of both Prokaryote and Eukaryotes: Cell Membrane, Genetic Material, Cytoplasm; Characteristic structural features of Prokaryotes; Characteristic structural features of Eukaryotes.	70 min



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	Cell structure and function (Part 2): Cell: Structural organization of bacteria showing major organelles; Structural organization of animal cell showing major organelles; Structural organization of plant cell showing major organelles; Differences between prokaryotes and eukaryotes; Name of organelles found in Eukaryotes	
	Interactive Q & A session	10 min
	Total	90 min
12	Recap and review on previous lecture	10 min
	Cell structure and function (Part 3): Structure and Functions of Cell membrane, Nucleus, Mitochondria, Chloroplasts, Ribosomes, Endoplasmic Reticulum, Golgi apparatus, Lysosomes, Peroxisomes. Difference between Plant and Animal cell structure.	70 min
	Interactive Q & A session	10 min
	Total	90 min
13	Midterm 1 If an examination is not held for unavoidable circumstances, it will be held on the following class day.	60 min
14	Recap and review on previous lecture	10 min
	Cellular Reproduction (Part 1): Cell cycle: Definition, Name of Different phases of cell cycle, What happens in each phase. Cell reproduction (Part 2): Definition, Types of cellular production; Asexual reproduction: Definition, Names of asexual production in animal, plant and bacteria with examples.	70 min
	Interactive Q & A session	10 min
	Total	90 min
15	Recap and review on previous lecture	10 min
	Cell Division (Part 3): Definition, Types of cell division; Mitosis: Characteristic features of mitosis, Overview of mitosis, Instances of Mitosis; Meiosis: Characteristic features of Meiosis, Overview of Meiosis, Instances of Meiosis; Number of Chromosomes in Human, Autosomes, Sex Chromosome; Somatic cell or Body cells, Sex cells or Gametes; Name of Organs that produces Gametes in Human and Plants; Zygote; Definition; Difference between mitosis and meiosis	70 min
	Interactive Q & A session	10 min
	Total	90 min
16	Recap and review on previous lecture	10 min
	Energy of Life (Part 1): Metabolism: Definition, Types of metabolism with examples, specific functions of metabolism; Aerobic and Anaerobic respiration with examples; Circumstances of anaerobic respiration with examples; Enzymes: Definition, Characteristics of enzymes;	70 min
	Interactive Q & A session	10 min



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	Total	90 min
17	Recap and review on previous lecture	10 min
	Energy of Life (Part 2): Cellular respiration: types of cellular respiration; Equation of aerobic respiration, where it occurs; Equation of anaerobic respiration by bacteria, What is it called; Equation of anaerobic respiration by Yeast; Instances anaerobic respiration occurs in animals including human, Equation for that; Photosynthesis: Respiration by Plant, Definition, Complete Equation, Role of Chloroplast.	70 min
	Interactive Q & A session	10 min
	Total	90 min
18-20	Recap and review on previous lecture	10*3 min
	Human Physiology: Homeostasis: Definition; Name organs involved in homeostasis; Basic mechanism by which homeostasis is maintained-Feedback mechanism; Role of thermoregulatory center (TRC); Name of regions of the hypothalamus of the brain responsible for the maintenance of different homeostatic, autonomic functions; Core body temperature: how it can vary normal ways; how temperature variation is retained to normal: regulation steps; What happens when we get too hot; What happens when we get too cold; Responding organ: Skin; Name three primary layers of skin; Functions of skin; Condition when the variation of normal core body temperature cannot be corrected: Fever and hyperthermia. Circulatory system (Part 1): Definition; Components; How it works; Blood: Definition; Function; Composition; Functions of blood cells and fluid part; Blood cells count how related with functional disorders or diagnosis; knowing Blood groups. Circulatory system (Part 2): Blood Vessels: Definition; Types; Difference between Artery and Veins. Heart: Components; Function; Illustration on how heart pumps blood and discuss; Illustration and discussion on heart disease and stroke can be developed; Preventive measures to avoid heart disease in terms of life and food styles. Respiratory system: Definition; Two Phases of respiration; Components; Illustration and discussion on how red blood cell carries oxygen from lung to tissue and tissue to lung. Excretory system: Definition; components with figure; functional unit of kidney; Osmoregulation; Functions of kidneys.	70*3 min
	Interactive Q & A session	10*3 min
	Total	90*3 min
21-23	.	10*3 min



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	Human Physiology: Digestive system: Digestion definition; components of Human Digestive System; different parts of human alimentary canal; Saliva, and its components; Role of Oral Cavity, Esophagus in digestion; Peristalsis; functions of Stomach; Chyme; functions of small Intestine ; Name and definition of accessory organs; Bile and role of bile in digestion; role of liver and pancreas. Health and Diseases: Food & Nutrition: Food: Definition, Function, Classification with examples; Name of basic nutrients requirement of human body; Unit of energy, Calorie value of Carbohydrates, Fats and Proteins; Balanced diet: Definition, Components; Name factors to be considered to formulate a balanced diet; Illustration of balanced diet chart for adult, total calorie requirements for adult Male and Female; BMI: Definition, Equation, BMI range for underweight, normal, overweight and obese: Significance of BMI; Vitamins: Definition, Classification, One deficiency disorder of each vitamins and food sources; Minerals: Important minerals of life: Sources, deficiency disorder, Malnutrition	70*3 min
	Interactive Q & A session	10*3 min
	Total	90*3 min
24		10 min
	Health and Diseases: Diabetes and Lipid Profile: Gestational diabetes: Definition, how it develops, possible fates of a baby born to mother having diabetes during pregnancy; Sign and symptoms of Diabetes; Diabetes diagnosis, Normal blood glucose level, glucose level considered as diabetic; Lipid profile: Definition, Tests included in lipid Profile, Implication of doing Lipid Profile test	70 min
	Interactive Q & A session and review (if needed)	10 min
	Total	90 min
	Final Exam: Lectures 14 – 24	75-90 min

**Lecture number may change depending on the situation and content covered.*

Textbooks:

- Kathleen Anne Ireland. Visualizing Human Biology 3rd Edition. John Wiley & Sons, Inc. 2011
- Reece Campbell Biology, 9th Edition.

Performance Evaluation:

The performance of a student will be evaluated according to the following criteria:

- Formative (Quizzes, Class participation, Presentation, Assignment etc.)



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ii) Summative (Mid-term and Final examination)

Examination	Marks	Remarks
Quizzes** (3)	20 (20%)	Four quizzes each carrying 10 marks. Quizzes will be objective type in a format that will include fill-in-the-blanks, true/false, MCQ and SAQ.
Class Attendance and participation/presentation*	5+5 (10%)	100% attendance is required, and there will be no make-up lecture for those absent. I shall render some help to students who miss a lecture for a serious reason.
Mid-Term Examination	30 (30%)	The Mid-Term Examination will be based on written questions and answers.
Final Examination	40 (40%)	The final Exam will be based on questions with answers in both essay format and short. There will be one Final Exam at the end of the course.
Total Marks	100	The grade will be prepared accordingly.

*When the presentation is taken, 5% marks will be taken from the summative assessment.

**There will be no 'make-up' exam for the quiz. (The percentage of distribution of all the assessment items and also the type of assessment items are subject to change, and any such change will be notified to the student ASAP)

*** If the class is shifted to online then the evaluation of midterm and final examinations may be changed to another mode of assessment like assignments and so on as per the university decided.

Grading scheme:

NSU grading policy will be followed <http://www.northsouth.edu/academic/grading-policy.html>.

Code of Conduct for students:

- **Class attendance:** 100% attendance is expected, and absences will detract from one's grade.
- **Class participation:** Intelligent, vocal participation in class will be a primary determinant of class participation grade, although other types of contributions will also be recognized.
- **Scheduling and Deadlines:** Please plan carefully. Exam schedules and deadlines for submitting assigned work are firm, and extensions will not be granted for conflicts with the workload for other courses. Accommodations due to illness or other personal situation must be requested and negotiated under special circumstances. Strictly follow the following please:
 1. No make-up quiz under any circumstance.
 2. Make-up for mid-term exams will be considered only under special circumstances under the prior approval from the proper authority. Any make-up of the mid-term exam will be based on questions with answers on essay format.
 3. If an exam is not held for unavoidable circumstances, it will be held on the following class day.
- **Academic Honesty:** Do not lie, cheat, copy or plagiarize. Do your own work on examinations. Please, read the policy about probation in NSU.



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Switch your mobile phones to vibration mode inside the class. Do not receive or make calls using your phone inside the class. Do not exchange messages using your phone inside the class.

Recording of class materials/presentations is not allowed under any circumstances and the use of pencils is also strictly prohibited during exams.

“Stay positive and happy. Work hard and don't give up hope. Be open to criticism and keep learning. Surround yourself with happy, warm, and genuine people”.

Tena Desae